AI for Social Research

2026 Masterclass in Social Research

Plan for today

- Motivation and objectives
- The "magic" and reality
- AI for social sciences
- Looking ahead

Motivation and objectives

- A moment of existential crisis? AI is reshaping how "knowledge" is produced
- Competing narratives: hype versus skepticism – both contain truth
- Social sciences face a specific challenge: AI both *is* a social phenomenon and *transforms* how we study social phenomena
- Goal: develop an informed, critical, and productive relationship with (gen-)AI tools

What Makes LLMs Groundbreaking: Statistical Learning at Scale

- **Statistical learning** – LLMs learn probability distributions over language: predicting the next token given context, with no hand-crafted rules or symbolic representations
- **Scaling laws** – model capability improves *predictably* with compute, parameters and data (Kaplan et al. 2020); intelligence as an emergent property of scale rather than explicit programming
- A huge amount of **training data** – models trained on internet-scale corpora absorb linguistic patterns, factual associations, and implicit cultural knowledge across domains and languages
- **Hardware** – GPU parallelism makes billion-parameter training feasible; compute constraints now shape the geopolitics of AI (chip export controls, sovereign AI infrastructure)

How do LLMs actually work?

- At its core, an LLM is a **next-token predictor** – given a sequence of words, it estimates the most probable continuation
- This is not retrieval (looking something up) and not reasoning (following logical rules) – it is **pattern completion** learned from statistical regularities in text
- The surprising result: predicting text at sufficient scale produces behaviour that *looks like* understanding, translation, coding, and argumentation – without any of these being explicitly programmed
- Implication for researchers: the model has no beliefs, intentions, or knowledge – only **learned associations between tokens**

Training data: The world encoded as text

- LLMs are trained on internet-scale corpora: web pages, books, code, academic papers, social media, legal documents
- The model does not *store* this text – it **compresses** it into billions of numerical parameters (weights) that encode statistical relationships between concepts
- What the model "knows" reflects what was *written down, in languages represented online, by people with internet access* – a deeply non-representative sample of human knowledge
- **Knowledge cutoff**: the model has no awareness of events after its training data ends – it cannot update in real time (without retrieval augmentation)

Scaling laws and emergent behaviour

- Kaplan et al. (2020) showed that LLM performance improves *predictably* with three factors: model size (parameters), training data volume, and compute
- More surprising: certain capabilities – multi-step reasoning, language translation, analogical thinking – **emerge abruptly** at scale thresholds, without being explicitly trained
- This makes frontier LLMs qualitatively different from earlier ML tools: their capabilities are **difficult to anticipate in advance** and difficult to fully audit after the fact
- For social researchers: you are working with a tool whose full capability envelope is not yet known – even to its developers

The Hallucination Problem: Fluent Does Not Mean True

- Because LLMs predict *plausible* text rather than *true* text, they routinely produce confident, fluent, **factually wrong outputs** – including fabricated citations, invented statistics, and non-existent court cases
- This is not a bug to be fixed – it is a **structural feature** of how the models work; accuracy is not what they are optimised for
- The danger for researchers: errors are *difficult to detect* precisely because they are stylistically indistinguishable from correct outputs
- Practical rule: **never use an LLM output as a primary source**; always verify factual claims, citations, and quotations independently

How the use of AI has changed: From GPT to AGI

- **Chatbot and API interface** – "conversational" AI; direct prompt–response interaction
- **Prompt engineering and context** – structured prompting, chain-of-thought, and retrieval-augmented generation (RAG)
- **Fine-tuning domain-specific models** – adapting pretrained LLMs to specialized corpora and tasks
- **Agentic AI and automated pipelines** – multi-step reasoning, tool use, and autonomous task execution
- **Agentic frameworks and harness engineering** – orchestrating agents at scale (e.g., OpenClaw, Claude Skills)

LLMs and social sciences: Three domains

- **Applied uses** – AI augmenting professional practice, such as creative arts, education (teachers and students), newsrooms, healthcare and counseling, and crime control
- **Human-centered AI** and **human-AI interactions** – AI as a force of structural change in human society, such as
  - Misinformation and democratic backsliding
  - Climate change and sustainable development
  - Labour market and economic development
- **Geopolitics/governance of AI** – development, regulation and governance, such as the Deepseek moment and US-China AI competition

AI as a research assistant: Practical starting points

- Identifying and refining research ideas, questions, and relevant literatures (e.g., "what has been done in this area?") – Hendriksen (2023)
- Locating potential co-authors and key/influential researchers in a field
- Stress-testing ideas via simulated critical readers (e.g., "how will my reviewers respond?")
- Drafting research applications, proposals, and code (with caution)

What social researchers can (should?) do with AI: Scale, scope, speed

- **Simulation** – explore human behaviour through AI "impersonation"; agent-based modeling (ABM) and human-machine comparisons
- **Data analysis and generation** – annotation, grounded-theory coding, NLP, discourse and thematic analysis (Geiecke & Jaravel 2024; Korinek 2023; Leek et al. 2024)
- **Survey and experimental research** – design, implementation and replication of surveys and experiments (Bail 2024)
- **Statistical programming** – "vibe coding" and beyond (more in Advanced Topics in Quantitative Social Research)

Bail

Caveats: What can go wrong

- **Human bias encoded at scale** – but also an opportunity to reverse-engineer how biases are (re)produced
- **Junk science and quality control** – "hallucination" and confabulation; domain-specific, context engineering, RAG and evaluation frameworks as partial solutions
- **Ethics as both problem and solution** – issues of deception, consent and simulation in AI-assisted research
- **Transparency and replicability** – performance evaluation, prompt versioning, and systematic model comparison

TOM CRUISE
A STEVEN SPIELBERG FILM
MINORITY REPORT
TWENTIETH CENTURY FOX AND DREAMWORKS PICTURES PRESENT A CRUISE/WAGNER/
BLUE TULIP/RONALD SHUSETT/GARY GOLDMAN PRODUCTION A STEVEN SPIELBERG FILM
TOM CRUISE "MINORITY REPORT" COLIN FARRELL SAMANTHA MORTON
AND MAX VON SYDOW JOHN WILLIAMS INDUSTRIAL LIGHT & MAGIC
DEBORAH L. SCOTT MICHAEL KAHN, A.C.E. ALEX McDOWELL
JANUSZ KAMINSKI, ASC GARY GOLDMAN RONALD SHUSETT
GERALD R. MOLEN BONNIE CURTIS WALTER F. PARKES JAN DE BONT
PHILIP K. DICK SCOTT FRANK AND JON COHEN STEVEN SPIELBERG
DREAMWORKS PICTURES
www.minorityreport.com
EVERYBODY RUNS JUNE 21

TECHNOLOGY

‘The Worst Internet-Research Ethics Violation I Have Ever Seen’

The most persuasive “people” on a popular subreddit turned out to be a front for a secret AI experiment.

By Tom Bartlett

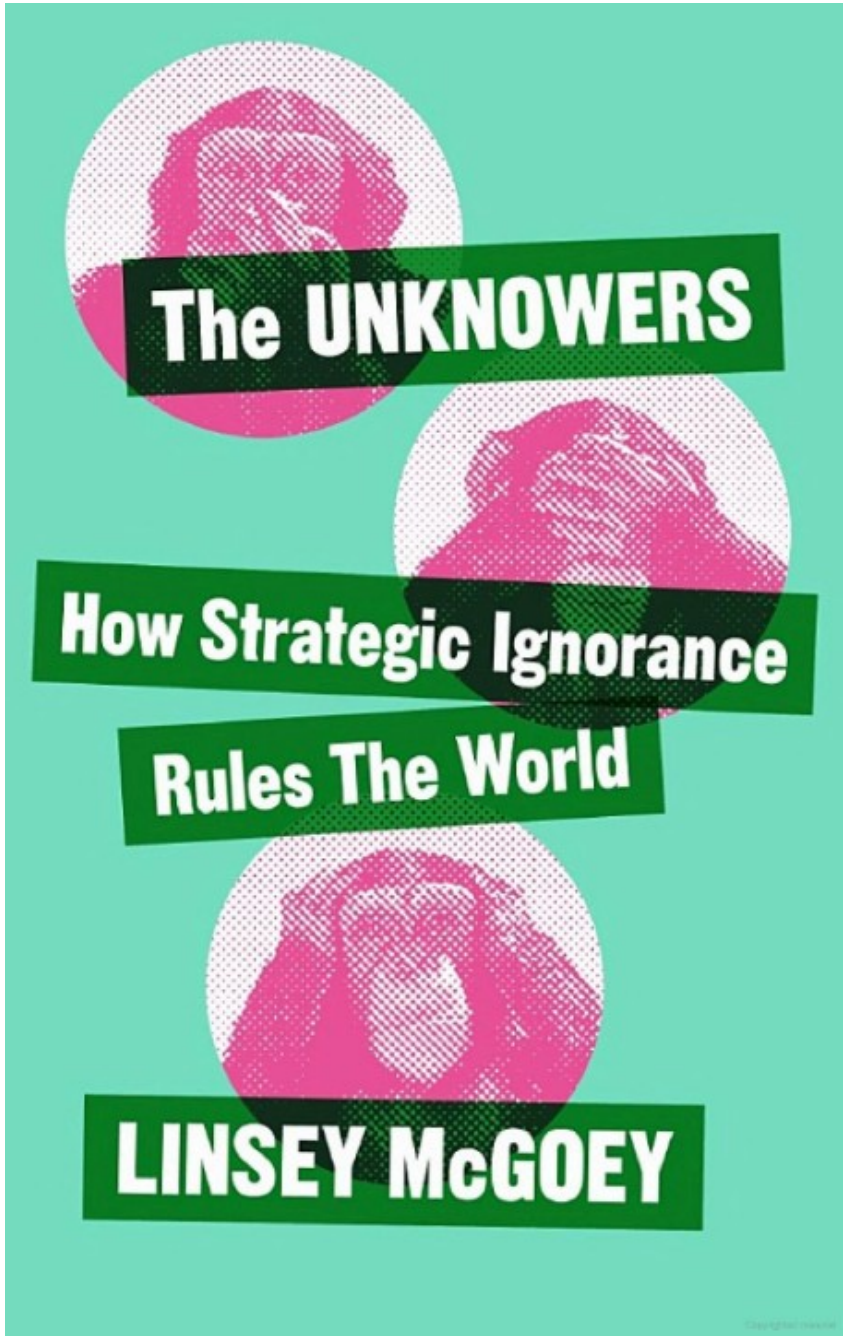
Illustration by The Atlantic

MAY 2, 2025

https://www.theatlantic.com/technology/archive/2025/05/reddit-ai-persuasion-experiment-ethics/682676/

How should researchers respond?

- Principle of **strategic ignorance** (McGoey 2019) – you do not need to understand AI fully to use it wisely; know enough to ask the right questions
- Identify the **skills that matter to you**: prompt/context/harness engineering, critical evaluation of outputs, understanding model limitations (at least at an intuitive level)
- Follow **institutional guidance**: Birkbeck and LSE policies on AI use and academic integrity


The UNKNOWERS
How Strategic Ignorance
Rules The World
LINSEY McGOEY

Questions for discussion

- Now that AI is a real possibility – how might *you* incorporate it into your research? How comfortable are you with using AI for research purposes, and why?
- Christopher Bail (PNAS 2024) identifies many promising research opportunities – which would you most like to explore? What opportunities does he miss?
- What skills do you need to acquire to take genuine advantage of AI? How much do you actually need to *know* about how AI works?

Online short course: "AI for Social Research" (in planning)

- **Session 1**: Overview – From GPT to AGI
- **Session 2**: LLMs and qualitative social research
- **Session 3**: LLMs for survey research
  - questionnaire design and pretesting
  - synthetic respondents and experiments
  - replication and validation
- **Sessions 4-6**: LLMs in social data science
  - natural language processing
  - agent-based modeling and simulations
  - agentic data analytical pipelines
- **Session 7**: Project workshop and looking ahead

Appendix: GenAI for plausibility checks

- Without using any conversational AI tools (e.g., ChatGPT, Claude and Gemini), imagine how you can integrate AI into your research practice and workflows
- Ask the tools you prefer to see what they can say/do for your research (both in terms of the topic and methods)
- Do these suggestions make sense to you? Any notable differences in terms of the quality of generated responses?